MATH 152 **EXAM I** Name_____

Mrs. Bonny Tighe 6.1-7.5

50 points

PART ONE TA_____Wed 10/2/2019

1. (11 points) Find the area of the region bounded the curves $y = \sin x, \ and \ y = \sin 2x$ on the interval $0 \le x \le \pi$. Graph.

2. (2 points each) True or false Circle the correct True or False

True False a) The functions $y = \tan^2 x$ and $y = \sec^2 x$ only vary by a constant.

True False b) $\int \frac{x^2+1}{x^2-4} dx = \int \frac{A}{x-2} + \frac{B}{x+2} dx$

True False c) $\int \frac{y^2}{\sqrt{4y^2-16}} dy$, we need to use the trig substitution $y = 2\sec\theta$.

True False d) The area between two curves could have a solution of $-\ln\left(\frac{2}{3}\right)$.

True False e) If $\tan\theta = \frac{(x+2)}{4}$, then $\sec\theta = \frac{\sqrt{(x+2)^2 - 16}}{4}$

True False f) To evaluate $\int \tan^2(w)dw$ you need to use *u*-substitution.

True False g) $\frac{1}{1-\cos x} = 1 - \sec x$.

True False h) $\int x^2 \sec(x^3)\,dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^2(x+1)}$ would be $\frac{Ax+B}{x^2} + \frac{C}{x+1}$

True False j) The formula for the average value of a function is $f_{avg} = \int_a^b f(x)dx$

3. (4 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^6 + x^4}$ have.

A) 7

B) 6

C) 1

D) 5

E) Some other answer

b) What method should not be used to integrate $\int \frac{x}{x^2 - 4}dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x) = 2x - 9x^2$ on the interval $[0, b]$ is -2, what is b ?

A) 3

B) 1

C) $3/2$

D) 0

E) Some other answer

d) After making a substitution for $\int \frac{x^2\,dx}{\sqrt{16+x^2}}$ you get

A) $\int 16\sin^2\theta\cos\theta\,d\theta$

B) $\int 16\tan^2\theta\sec\theta\,d\theta$

C) $\int 16\tan^2\theta\,d\theta$

D) $\int \frac{1}{\sqrt{u}}\,du$, where $u = 16 + x^2$

E) Some other answer

e) Using the Cylindrical Shells method to find the volume of the solid of revolution for the function $y = 4x - x^2$ about the line $x = 4$, you would have:

A) $2\pi \int_0^4 (4 - x)(4x - x^2 + 4)dx$

B) $2\pi \int_0^4 (x - 4)(4x - x^2)dx$

C) $2\pi \int_0^4 (4 - x)(4x - x^2)\, dx$

D) $2\pi \int_0^4 (x - 4)(4x - x^2 - 4)\, dx$

E) Some other answer

f) On which of the following integrals can you NOT use the method of partial fractions ?

A) $\int_0^1 \frac{1}{x^2 - 16} dx$

B) $\int_0^1 \frac{1}{x^3 + 16} dx$

C) $\int_0^1 \frac{1}{x^2 + 16} dx$

D) $\int_0^1 \frac{1}{x^4 - 16} dx$

E) You can use partial fractions for all of the above.

MATH 152 **EXAM I** NAME_____

Mrs. Bonny Tighe 6.1-7.5

50 points TA_____Wed10/2/2019

PART TWO

1. (10 points) Integrate $\int \frac{(x+2)^2}{\sqrt{5 - x^2 - 4x}} dx$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $x = 4y - y^2$ *and* $y = x$ rotated about the line y = 4. Graph

3. (10 points) Integrate: $\int \cos(\ln t)\, dt$

4. (10 points) Integrate: $\int_{\sqrt{3}}^{3} \frac{x^3}{\sqrt{9+x^2}}\, dx$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\, dx$

6. (5 points) Integrate $\int \tan^{-1} \sqrt{x}\, dx$

MATH 152 **EXAM IA** Name_____

Mrs. Bonny Tighe 6.1-7.5

50 points

PART ONE TA_____Wed 10/2/2019

1. (11 points) Find the area of the region bounded the curves $y = \sin 2x, \quad and \quad y = \sin x$ on the interval $0 \le x \le \frac{\pi}{2}$. Graph.

2. (2 points each) True or false Circle the correct True or False

True False a) The functions $y = \sin^2 x$ and $y = \cos^2 x$ only vary by a constant.

True False b) $\int \frac{x^2 + 1}{x^2 - 9} dx = \int \frac{A}{x - 3} + \frac{B}{x + 3} dx$

True False c) $\int \sec^4 y \tan^3 y \, dy$, can be integrated by using $u = \sec y$ and also $u = \tan y$.

True False d) The area between two curves could have a solution of $-\ln\left(\frac{3}{4}\right)$

True False e) If $\tan\theta = \frac{(x+2)}{4}$, then $\sec\theta = \frac{\sqrt{(x+2)^2 - 16}}{4}$

True False f) To evaluate $\int \cot^2(w)dw$ you need to use *u*-substitution.

True False g) $\frac{1}{1-\sin x} = 1 - \csc x$.

True False h) $\int x^2 e^{x^3} dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^3(x+1)}$ would be $\frac{Ax^2 + Bx + C}{x^3} + \frac{D}{x+1}$

True False j) The formula for the average value of a function is $f_{avg} = \int_a^b f(x)dx$

3. (4 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^5 + x^3}$ have.

A) 4

B) 6

C) 1

D) 5

E) Some other answer

b) What method should NOT be used to integrate $\int \frac{x}{x^2 - 4} dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x) = 4x - 9x^2$ on the interval $[0, b]$ is -1, what is b ?

A) 3

B) 1

C) $1/3$

D) 0

E) Some other answer

d) After making a substitution for $\int \frac{x^2\,dx}{\sqrt{16 - x^2}}$

A) $\int 4\sin^2\theta \; d\theta$

B) $\int \frac{4\sin^2\theta}{\cos\theta} d\theta$

C) $\int 16\sin^2\theta\, d\theta$

D) $\int \frac{1}{\sqrt{u}} du$, where $u = 16 - x^2$

E) Some other answer

e) Using the Cylindrical Shells method to find the volume of the solid of revolution for the function $y = 5x - x^2$ about the line $x = 5$, you would have:

A) $2\pi \int_0^4 (5 - x)(5x - x^2 + 5)dx$

B) $2\pi \int_0^4 (x - 5)(5x - x^2)\,dx$

C) $2\pi \int_0^4 (5 - x)(5x - x^2)\,dx$

D) $2\pi \int_0^4 (x - 5)(5x - x^2 - 5)\,dx$

E) Some other answer

f) On which of the following can you NOT use the method of partial fractions?

A) $\int_0^1 \frac{1}{x^2 - 16}\,dx$

B) $\int_0^1 \frac{1}{x^3 + 16}\,dx$

C) $\int_0^1 \frac{1}{x^2 + 16}\,dx$

D) $\int_0^1 \frac{1}{x^4 - 16}\,dx$

E) Some other answer

MATH 152
Mrs. Bonny Tighe

EXAM IA

6.1-7.5
50 points
PART TWO

NAME_____

TA_____Wed10/2/2019

1. (10 points) Integrate $\int \frac{(x+1)^2}{\sqrt{3 - x^2 - 2x}}\, dx$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $y = 6x - x^2$ *and* $y = x$ rotated about the line x = 6. Graph

3. (10 points) Integrate: $\int \sin(\ln t)\, dt$

4. (10 points) Integrate: $\int_3^6 \frac{x^3}{\sqrt{x^2 - 9}}\, dx$

5. (10 points) Integrate $\int \frac{x^5 + 2}{x^4 + x^2}\, dx$

6. (5 points) Integrate $\int \frac{e^x}{e^{2x}+1}\,dx$